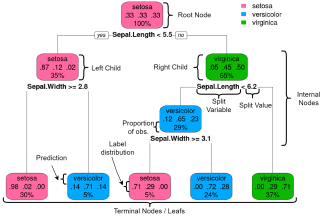
Introduction to Machine Learning

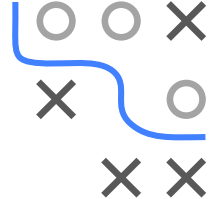
Advanced Risk Minimization
Loss functions and tree splitting
(Deep-Dive)



Learning goals

- Tree splitting loss vs impurity:
- Bernoulli loss $\sim$ entropy splitting
- Brier score $\sim$ gini splitting

RISK MINIMIZATION AND IMPURITY



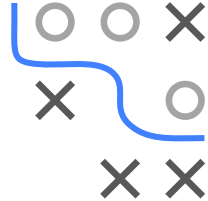
- Tree fitting: Find best way to split parent node $\mathcal{N}_0$ into child nodes $\mathcal{N}_1$ and $\mathcal{N}_2$ such that $\mathcal{N}_1 \cup \mathcal{N}_2 = \mathcal{N}_0$ and $\mathcal{N}_1 \cap \mathcal{N}_2 = \emptyset$
- Two options for evaluating how good a split is: Per node $\mathcal{N}$ compute the following:
  1. Compute impurity $\text{Imp}(\mathcal{N})$ directly from observations in $\mathcal{N}$
  2. Fit optimal constant using loss function, sum up losses for $\mathcal{N}$
- Two common impurity measures are entropy and Gini index:

$$\text{Imp}^{\text{ent}}(\mathcal{N}) = -\sum_{k=1}^{g} \pi_k^{(\mathcal{N})} \log \pi_k^{(\mathcal{N})}, \quad \text{Imp}^{\text{Gini}}(\mathcal{N}) = \sum_{k=1}^{g} \pi_k^{(\mathcal{N})}(1-\pi_k^{(\mathcal{N})})$$

- The impurity of a split is then calculated as a weighted average ($n_0 = n_1 + n_2$ are number of obs in nodes)

$$\text{Imp(split)} = \frac{n_1}{n_0}\text{Imp}(\mathcal{N}_1) + \frac{n_2}{n_0}\text{Imp}(\mathcal{N}_2) \quad (1)$$

RISK MINIMIZATION AND IMPURITY



- In the following we will prove that entropy and Gini impurity measures are equivalent to splitting using log loss and Brier score:

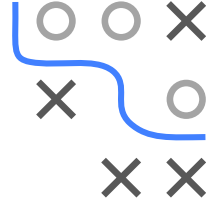
$$L(y, \pi(\mathbf{x})) = -\sum_{k=1}^{g} \mathbb{I}[y = k] \log(\pi_k(\mathbf{x})) \quad \text{(log-loss)}$$

$$L(y, \pi(\mathbf{x})) = \sum_{k=1}^{g} (\mathbb{I}[y = k] - \pi_k(\mathbf{x}))^2 \quad \text{(Brier)}$$

- Here, the empirical risk of a split is the unweighted sum of per-node risks:

$$\mathcal{R}_{\text{emp}}(\text{split}) = \mathcal{R}_{\text{emp}}(\mathcal{N}_1) + \mathcal{R}_{\text{emp}}(\mathcal{N}_2) \quad (2)$$

BERNOULLI LOSS MIN = ENTROPY SPLITTING



Claim: Using entropy as impurity

$$\text{Imp}(\mathcal{N}) = -\sum_{k=1}^{g} \pi_k^{(\mathcal{N})} \log \pi_k^{(\mathcal{N})} \quad \text{in (1)}$$

is equivalent to using (multiclass) Bernoulli loss

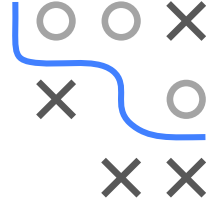
$$L\left(y, \pi(\mathbf{x})\right) = -\sum_{k=1}^{g} \mathbb{I}[y = k] \log(\pi_k(\mathbf{x})) \quad \text{in (2)}$$

Proof: Let $\mathcal{N} \subseteq \mathcal{D}$ denote the subset of observations in that node and consider $\mathcal{R}_{\text{emp}}(\mathcal{N})$ of node $\mathcal{N}$ with (multiclass) Bernoulli loss

$\Rightarrow$ Optimal constant per node $\pi_k^{(\mathcal{N})} = \frac{1}{n_{\mathcal{N}}} \sum_{(\mathbf{x},y)\in\mathcal{N}} \mathbb{I}[y = k] = \frac{n_{\mathcal{N},k}}{n_{\mathcal{N}}}$

where $n_{\mathcal{N},k}$ is the number of class k observations in node $\mathcal{N}$

RISK MINIMIZATION AND IMPURITY

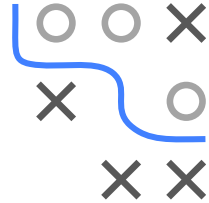


$$
\begin{aligned}
\mathcal{R}_{\text{emp}}(\mathcal{N}) &= \sum_{(\mathbf{x},y)\in\mathcal{N}} \left(-\sum_{k=1}^{g} \mathbb{I}[y=k] \log \pi_k(\mathbf{x})\right) \\
&= -\sum_{k=1}^{g} \sum_{(\mathbf{x},y)\in\mathcal{N}} \mathbb{I}[y=k] \log \pi_k^{(\mathcal{N})} \\
&= -\sum_{k=1}^{g} n_{\mathcal{N},k} \log \pi_k^{(\mathcal{N})} = -\sum_{k=1}^{g} (n_{\mathcal{N}} \cdot \pi_k^{(\mathcal{N})}) \log \pi_k^{(\mathcal{N})} \\
&= -\, n_{\mathcal{N}} \sum_{k=1}^{g} \pi_k^{(\mathcal{N})} \log \pi_k^{(\mathcal{N})} = n_{\mathcal{N}} \text{Imp}(\mathcal{N})
\end{aligned}
$$

$$
\begin{aligned}
\Rightarrow \mathcal{R}_{\text{emp}}(\text{split}) &= \mathcal{R}_{\text{emp}}(\mathcal{N}_1) + \mathcal{R}_{\text{emp}}(\mathcal{N}_2) = n_1 \text{Imp}(\mathcal{N}_1) + n_2 \text{Imp}(\mathcal{N}_2) \\
&= n_0 \left(\frac{n_1}{n_0} \text{Imp}(\mathcal{N}_1) + \frac{n_2}{n_0} \text{Imp}(\mathcal{N}_2)\right) = n_0 \text{Imp}(\text{split})
\end{aligned}
$$

Bernoulli-risk of the split $\mathcal{R}_{\text{emp}}(\text{split})$ is proportional to its entropy-impurity Imp(split), i.e., $\arg\min_{\text{split}} \mathcal{R}_{\text{emp}}(\text{split}) = \arg\min_{\text{split}} \text{Imp}(\text{split})$

BRIER SCORE MINIMIZATION = GINI SPLITTING



Claim: Using Gini as impurity

$$\text{Imp}(\mathcal{N}) = \sum_{k=1}^{g} \pi_k^{(\mathcal{N})}(1 - \pi_k^{(\mathcal{N})}) \quad \text{in (1)}$$

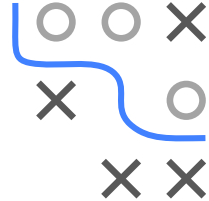
is equivalent to using Brier score

$$L\left(y, \pi(\mathbf{x})\right) = \sum_{k=1}^{g} (\mathbb{I}[y = k] - \pi_k(\mathbf{x}))^2 \quad \text{in (2)}$$

Proof: Empirical risk $\mathcal{R}_{\text{emp}}(\mathcal{N})$ of node $\mathcal{N}$ using (multiclass) Brier score has optimal constant per node:

$$\pi_k^{(\mathcal{N})} = \frac{1}{n_{\mathcal{N}}} \sum_{(\mathbf{x},y)\in\mathcal{N}} \mathbb{I}[y = k] = \frac{n_{\mathcal{N},k}}{n_{\mathcal{N}}}$$

BRIER SCORE MINIMIZATION = GINI SPLITTING



Inserting the optimal constant, the risk simplifies to

$$\mathcal{R}_{\text{emp}}(\mathcal{N}) = \sum_{(\mathbf{x},y)\in\mathcal{N}} \sum_{k=1}^{g} (\mathbb{I}[y=k] - \pi_k^{(\mathcal{N})})^2 = \sum_{k=1}^{g} \sum_{(\mathbf{x},y)\in\mathcal{N}} (\mathbb{I}[y=k] - \frac{n_{\mathcal{N},k}}{n_{\mathcal{N}}})^2$$

$$= \sum_{k=1}^{g} \Big(\sum_{(\mathbf{x},y)\in\mathcal{N}:\ y=k} (1 - \frac{n_{\mathcal{N},k}}{n_{\mathcal{N}}})^2 + \sum_{(\mathbf{x},y)\in\mathcal{N}:\ y\neq k} (0 - \frac{n_{\mathcal{N},k}}{n_{\mathcal{N}}})^2 \Big)$$

$$= \sum_{k=1}^{g} n_{\mathcal{N},k} (1 - \frac{n_{\mathcal{N},k}}{n_{\mathcal{N}}})^2 + (n_{\mathcal{N}} - n_{\mathcal{N},k}) (\frac{n_{\mathcal{N},k}}{n_{\mathcal{N}}})^2$$

since for $n_{\mathcal{N},k}$ observations the condition $y = k$ is met, and for the remaining $(n_{\mathcal{N}} - n_{\mathcal{N},k})$ observations it is not.

BRIER SCORE MINIMIZATION = GINI SPLITTING

We further simplify the expression to

$$
\begin{aligned}
\mathcal{R}_{\text{emp}}(\mathcal{N}) &= \sum_{k=1}^{g} n_{\mathcal{N},k}\left(\frac{n_{\mathcal{N}} - n_{\mathcal{N},k}}{n_{\mathcal{N}}}\right)^2 + (n_{\mathcal{N}} - n_{\mathcal{N},k})\left(\frac{n_{\mathcal{N},k}}{n_{\mathcal{N}}}\right)^2 \\
&= \sum_{k=1}^{g} \frac{n_{\mathcal{N},k}}{n_{\mathcal{N}}} \frac{n_{\mathcal{N}} - n_{\mathcal{N},k}}{n_{\mathcal{N}}} (n_{\mathcal{N}} - n_{\mathcal{N},k} + n_{\mathcal{N},k}) \\
&= n_{\mathcal{N}} \sum_{k=1}^{g} \pi_k^{(\mathcal{N})} \cdot (1 - \pi_k^{(\mathcal{N})}) = n_{\mathcal{N}} \text{Imp}(\mathcal{N}) \\
\Rightarrow \mathcal{R}_{\text{emp}}(\text{split}) &= \mathcal{R}_{\text{emp}}(\mathcal{N}_1) + \mathcal{R}_{\text{emp}}(\mathcal{N}_2) = n_1 \text{Imp}(\mathcal{N}_1) + n_2 \text{Imp}(\mathcal{N}_2) \\
&= n_0\left(\frac{n_1}{n_0}\text{Imp}(\mathcal{N}_1) + \frac{n_2}{n_0}\text{Imp}(\mathcal{N}_2)\right) = n_0 \text{Imp}(\text{split})
\end{aligned}
$$

Brier-risk $\mathcal{R}_{\text{emp}}(\text{split})$ of the split is proportional to its gini-impurity Imp(split), i.e., $\arg\min_{\text{split}} \mathcal{R}_{\text{emp}}(\text{split}) = \arg\min_{\text{split}} \text{Imp}(\text{split})$